arXiv:2412.14171v2 [cs.CV] 2 Jul 2025

Thinking in Space: How Multimodal Large Language Models See, Remember, and Recall Spaces

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Project Page Evaluation Code VSI-Bench

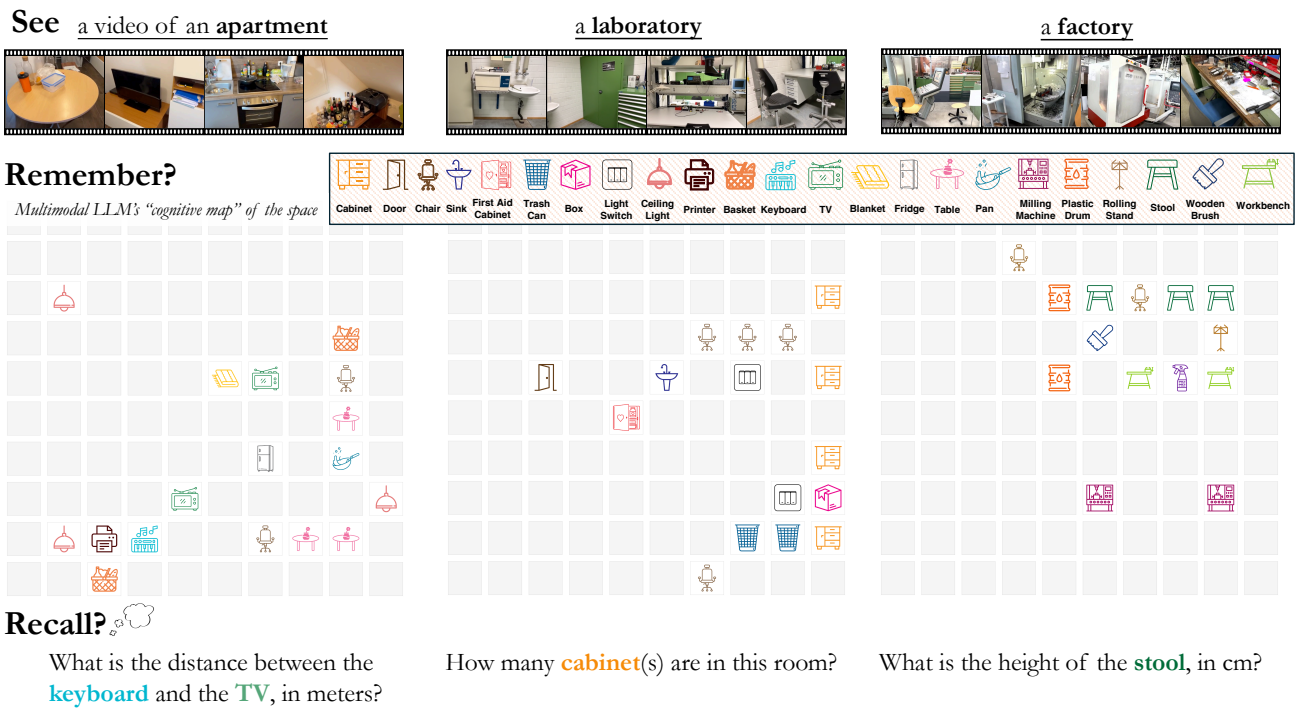


Figure 1. Whether at home, in the workplace, or elsewhere, the ability to perceive a space, remember its layout, and retrieve this spatial information to answer questions on demand is a key aspect of visual-spatial intelligence. Recent Multimodal LLMs can understand general videos, but can they "think spatially" when presented with a video recording of an environment? Can they build an accurate, implicit "*cognitive map*" that allows them to answer questions about a space? What are the strengths and limitations of using MLLMs to enhance spatial intelligence? We dig into these questions by setting up video data for MLLMs to watch, building a VQA benchmark to check their recall, and examining what the MLLMs actually remember and understand.

Abstract

*Humans possess the visual-spatial intelligence to remember spaces from sequential visual observations. However, can Multimodal Large Language Models (MLLMs) trained on million-scale video datasets also "think in space" from videos? We present a novel video-based visual-spatial intelligence benchmark (VSI-Bench) of over 5,000 question-answer pairs, and find that MLLMs exhibit competitive—though subhuman—visual-spatial intelligence. We probe models to express how they think in space both linguistically and visually and find that while spatial reasoning capabilities remain the primary bottleneck for MLLMs to reach higher benchmark performance, local world models and spatial awareness do emerge within these models. Notably, prevailing linguistic reasoning techniques (*e.g.*, chain-of-thought, self-consistency, tree-of-thoughts) fail to improve performance, whereas explicitly generating cognitive maps during question-answering enhances MLLMs' spatial distance ability.*

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1. Introduction

When shopping for furniture, we often try to recall our living room to imagine if a desired cabinet will fit. Estimating distances is difficult, yet after even a single viewing, humans can mentally reconstruct spaces, recalling objects in a room, their positions, and sizes. We live in a sensory-rich 3D world where visual signals surround and ground us, allowing us to perceive, understand, and interact with it.

Visual-spatial intelligence entails perceiving and mentally manipulating spatial relationships [26]; it requires myriad capabilities, including relational reasoning and the ability to transform between egocentric and allocentric perspectives (Sec. 2). While Large Language Models (LLMs) [3, 6, 9, 36, 61, 67, 68, 77, 81, 82, 88, 105] have advanced linguistic intelligence, visual-spatial intelligence remains under-explored, despite its relevance to robotics [7, 8, 21, 64], autonomous driving [79], and AR/VR [12, 27, 55].

Multimodal Large Language Models (MLLMs) [1, 4, 15, 34, 42, 49, 49, 78], which integrate language and vision, exhibit strong capacities to think and reason in open-ended dialog and practical tasks like web agents [21, 28, 33, 35]. To advance this intelligence in the visual-spatial realm, we introduce `VSI-Bench`, a video-based benchmark featuring over 5,000 question-answer pairs across nearly 290 real indoor-scene videos (Sec. 3). Video data, by capturing continuous, temporal input, both parallels how we observe the world and enables richer spatial understanding and reasoning than static images. Evaluating open- and closed-source models on `VSI-Bench` reveals that even though a large performance gap exists between models and humans, MLLMs exhibit emerging visual-spatial intelligence despite the challenges of video understanding, textual understanding, and spatial reasoning (Sec. 4).

To analyze model behavior and inspired by dual-coding theory [18], which posits that linguistic and visual processing are distinct yet complementary, we prompt selected models for self-explanations (linguistic) and cognitive maps (visual). Analyzing the self-explanations reveals that spatial reasoning, as compared to visual perception, linguistic intelligence, or temporal processing, is the main factor behind weak performance on `VSI-Bench` (Sec. 5). "*Cognitive maps*", which represent internal layouts of environments [62, 80], allow us to evaluate MLLMs' implicit spatial world models and find that MLLMs build strong local models but weak global ones (Sec. 6). Furthermore, standard linguistic reasoning techniques fail to enhance performance on our benchmark. However, explicitly generating and using cognitive maps improves spatial distance question-answering.

Expressing visual-spatial intelligence is difficult (and often piecemeal), even for humans [26]. With this work, we aim to encourage the community to explore grounding frontier models with visual-spatial intelligence and to pave and illuminate this direction.

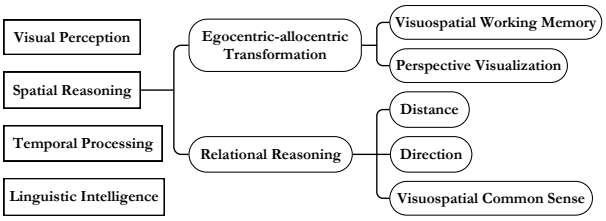


Figure 2. A taxonomy of **visual-spatial intelligence** capabilities.

2. Visual-Spatial Intelligence

We discuss preliminaries and scope visual-spatial intelligence to provide context and a framework for later analysis.

Term Use. We use "intelligence" rather than "cognition" as it is broader, and "spatial cognition" is a branch of cognitive psychology [83]. We prefix spatial intelligence in our work with "visual", as spatial intelligence exists irrespective of sensory modality (*e.g.*, a blind person can perceive space through other senses) [26]. Given our focus on video input, we discuss *visual*-spatial intelligence.

Investigation Scope. While classic spatial intelligence tests also include pen-paper tasks like the Mental Rotation Test [74], our focus is on visual-spatial intelligence as it applies to real-world environments, particularly in common spaces like homes, offices, and factories.

Taxonomy. We provide a taxonomy of capabilities potentially required for visual-spatial intelligence (Fig. 2), based on cognitive psychology [11, 26, 57, 62] and human experience with our benchmark tasks in Sec. 3. Visual perception, linguistic intelligence, temporal processing, and spatial reasoning are the four areas needed in `VSI-Bench`. For example, [11] shows that visual object and spatial processing are neurally distinct, which motivates "visual perception" and "spatial reasoning" as separate areas. We break spatial reasoning into two broad capabilities: relational reasoning and egocentric-allocentric transformation.

Relational reasoning is the ability to identify, via distance and direction, relationships between objects. It also encompasses reasoning about distance between objects by relying on visuospatial common sense about the sizes of other objects. For example, knowing a standard beverage can is approximately 12 cm tall, humans can estimate other object sizes by comparing visual proportions.

Egocentric-allocentric transformation involves shifting between a self-centered (egocentric) view and an environment-centered (allocentric) one. In our setting, each egocentric video frame maps to allocentric object positions and camera trajectory. When humans observe a space, they convert egocentric perceptions into an allocentric mental map, enabling perspective-taking from various viewpoints—essential for tasks like relative direction or route planning. This transformation relies on visualizing new perspectives and on visuospatial working memory [2], the abil-



Figure 3. **Tasks demonstration of VSI-Bench**. Note: the questions above are simplified slightly for clarity and brevity.

ity to hold and manipulate spatial information, say by updating object positions from new egocentric input [20, 56].

Every task in VSI-Bench requires perceptual, linguistic, and temporal abilities and varying degrees of spatial reasoning. For example, egocentric-allocentric transformation is much more important for a task like route planning than object size estimation. These factors provide some context for the complexity of visual-spatial intelligence.

3. VSI-Bench

3.1. Overview

We introduce VSI-Bench to quantitatively evaluate the visual-spatial intelligence of MLLMs from egocentric video. VSI-Bench comprises over 5,000 question-answer pairs derived from 288 real videos. These videos are sourced from the validation sets of the public indoor 3D scene reconstruction datasets ScanNet [19], ScanNet++ [97], and ARKitScenes [5] and represent diverse environments—including residential spaces, professional settings (*e.g.*, offices, labs), and industrial spaces (*e.g.*, factories)—and multiple geographic regions. Repurposing these existing 3D reconstruction and understanding datasets offers accurate object-level annotations which we use in question generation and could enable future study into the connection between MLLMs and 3D reconstruction. VSI-Bench is high-quality, having been iteratively reviewed to minimize question ambiguity and to remove incorrect annotations propagated from the source datasets.

VSI-Bench includes eight tasks of three types: *configurational*, *measurement estimation*, and *spatiotemporal*. The configurational tasks (*object count, relative distance, relative direction, route plan*) test a model's understanding of the configuration of a space and are more intuitive for humans (see Sec. 4 for comparison between MLLM and human performance). Measurement estimation (of *object size, room size*, and *absolute distance*) is of value to any embodied agent. While predicting a measurement exactly is very difficult, for both humans and models, a better sense of distance and other measurements is intuitively correlated with better visual-spatial intelligence and underpins a wide range of tasks that require spatial awareness, like interaction with objects and navigation. Spatiotemporal tasks like *appearance order* test a model's memory of a space as seen in video. See Fig. 3 for an overview of VSI-Bench tasks and Fig. 5 for dataset statistics.

3.2. Benchmark Construction

We develop a sophisticated benchmark construction pipeline to effectively generate high-quality question-answer (QA) pairs at scale, as shown in Fig. 4.

Data Collection and Unification. We begin our dataset construction by standardizing various datasets into a unified meta-information structure, ensuring dataset-agnostic QA pair generation. Our benchmark aggregates existing 3D indoor scene understanding and reconstruction datasets: ScanNet [19], ScanNet++ [97], and ARKitScenes [5]. These datasets provide high-fidelity video scans capable of space reconstruction, ensuring MLLMs can answer space-level questions with only video input. Additionally, their object-level 3D annotations facilitated our question generation. We parse the datasets into a unified meta-information format including object categories, bounding boxes, video specifications (resolution and frame rate), and more.

Question-Answer Generation. QA pairs are primarily auto-annotated using the meta-information and question templates; the *route plan* task was human-annotated. We sophisticatedly design and refine the question template for each task and provide guidelines for human annotators. For

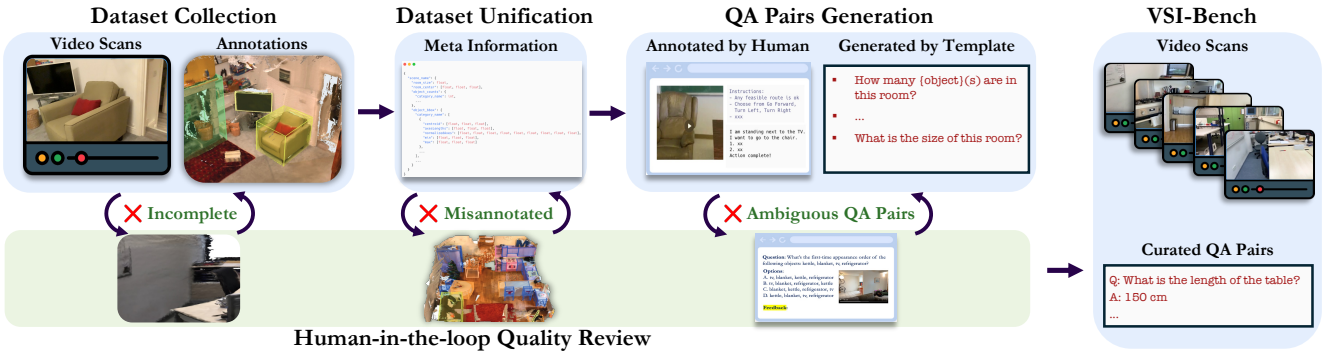


Figure 4. **Benchmark curation pipeline.** The pipeline first unifies diverse datasets into a standardized format and semantic space for consistent processing. QA pairs are then generated through both human annotation and question templates. To ensure quality, human verification is implemented at all key stages for filtering low-quality videos, annotations, and ambiguous QA pairs.

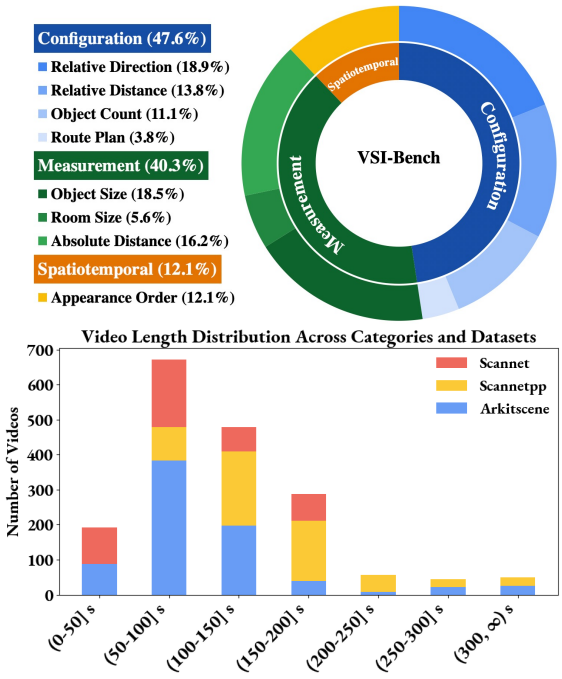


Figure 5. **Benchmark Statistics**. **Top**: The distribution of tasks across three main categories. **Bottom**: The video length statistic.

more detailed design, see Appendix B.1.

Human-in-the-loop Quality Review. Despite human-annotated data sources and a meticulously designed QA generation methodology, certain ambiguities and errors inevitably persist, primarily due to inherent annotation errors in the source datasets. We implement a human-in-the-loop verification protocol spanning benchmark construction. This iterative quality assurance is bidirectional: when evaluators flag ambiguous or erroneous questions, we trace the error source and remove the problematic data sample or modify the meta-information, question template, or QA generation rule accordingly to rectify other erroneous questions stemming from the same source. Following each human review cycle, we update and iterate the benchmark until it satisfies our quality standards.

4. Evaluation on VSI-Bench

4.1. Evaluation Setup

Benchmark Models. We comprehensively evaluate 15 video-supporting MLLMs across diverse model families, encompassing various parameter scales and training recipes. For proprietary models, we consider Gemini-1.5 [78] and GPT-4o [34]. For open-source models, we evaluate models from InternVL2 [14], ViLA [45], LongViLA [91], LongVA [101], LLaVA-OneVision [40], and LLaVA-Video [104]. All evaluations are conducted under zero-shot settings and using each model's default prompts. To ensure reproducibility, we use greedy decoding for all models.

Metric Design. Based on whether the ground-truth answer is verbal or numerical, our tasks are suited to either a Multiple-Choice Answer (MCA) or Numerical Answer (NA) format (see Fig. 3). For MCA tasks, we follow standard practice [24, 31, 99] by using *Accuracy* ($\mathcal{ACC}$), based on exact matching (with possible fuzzy matching), as the primary metric. For NA tasks, where models predict continuous values, accuracy via exact matching fails to capture the degree of proximity between model predictions and ground-truth answers. Therefore, we introduce a new metric, *Mean Relative Accuracy* ($\mathcal{MRA}$), inspired by previous works [22, 46, 73]. Specifically, for a NA question, given a model's prediction $\hat{y}$, ground truth y, and a confidence threshold θ, relative accuracy is calculated by considering $\hat{y}$ correct if the relative error rate, defined as $|\hat{y}-y|/y$, is less than $1-\theta$. As single-confidence-threshold accuracy only considers relative error in a narrow scope, $\mathcal{MRA}$ averages the relative accuracy across a range of confidence thresholds $\mathcal{C} = \{0.5, 0.55, \dots, 0.95\}$:

$$\mathcal{MRA} = \frac{1}{10}\sum_{\theta\in\mathcal{C}} \mathbb{1}\left(\frac{|\hat{y}-y|}{y} < 1-\theta\right). \quad (1)$$

$\mathcal{MRA}$ offers a more reliable and discriminative measurement for calculating the similarity between numerical predictions and ground truth values.

Chance Level Baselines. We provide two baselines:

Methods	Rank	Avg.	Obj. Count	Abs. Dist.	Obj. Size	Room Size	Rel. Dist.	Rel. Dir.	Route Plan	Appr. Order
			Numerical Answer				Multiple-Choice Answer			
Baseline										
Chance Level (Random)	-	-	-	-	-	-	25.0	36.1	28.3	25.0
Chance Level (Frequency)	-	34.0	62.1	32.0	29.9	33.1	25.1	47.9	28.4	25.2
VSI-Bench (tiny) Perf.										
†Human Level	-	79.2	94.3	47.0	60.4	45.9	94.7	95.8	95.8	100.0
†Gemini-1.5 Flash	-	45.7	50.8	33.6	56.5	45.2	48.0	39.8	32.7	59.2
†Gemini-1.5 Pro	-	48.8	49.6	28.8	58.6	49.4	46.0	48.1	42.0	68.0
†Gemini-2.0 Flash	-	45.4	52.4	30.6	66.7	31.8	56.0	46.3	24.5	55.1
Proprietary Models (API)										
GPT-4o	3	34.0	46.2	5.3	43.8	38.2	37.0	41.3	31.5	28.5
Gemini-1.5 Flash	2	42.1	49.8	30.8	53.5	54.4	37.7	41.0	31.5	37.8
Gemini-1.5 Pro	1	45.4	56.2	30.9	64.1	43.6	51.3	46.3	36.0	34.6
Open-source Models										
InternVL2-2B	11	26.5	25.7	24.0	20.0	29.2	32.1	44.1	30.4	6.3
InternVL2-8B	3	37.5	31.3	29.0	48.9	44.2	38.0	33.4	28.9	46.4
InternVL2-40B	4	37.0	41.3	26.2	48.2	27.5	47.6	32.7	27.8	44.7
LongVILA-8B	12	21.6	29.1	9.1	16.7	0.0	29.6	30.7	32.5	25.5
VILA-1.5-8B	9	28.9	17.4	21.8	50.3	18.8	32.1	34.8	31.0	24.8
VILA-1.5-40B	7	31.2	22.4	24.8	48.7	22.7	40.5	25.7	31.5	32.9
LongVA-7B	8	29.2	38.0	16.6	38.9	22.2	33.1	43.3	25.4	15.7
LLaVA-Video-7B	5	35.6	48.5	14.0	47.8	24.2	43.5	42.4	34.0	30.6
LLaVA-Video-72B	1	40.9	48.9	22.8	57.4	35.3	42.4	36.7	35.0	48.6
LLaVA-OneVision-0.5B	10	28.0	46.1	28.4	15.4	28.3	28.9	36.9	34.5	5.8
LLaVA-OneVision-7B	6	32.4	47.7	20.2	47.4	12.3	42.5	35.2	29.4	24.4
LLaVA-OneVision-72B	2	40.2	43.5	23.9	57.6	37.5	42.5	39.9	32.5	44.6

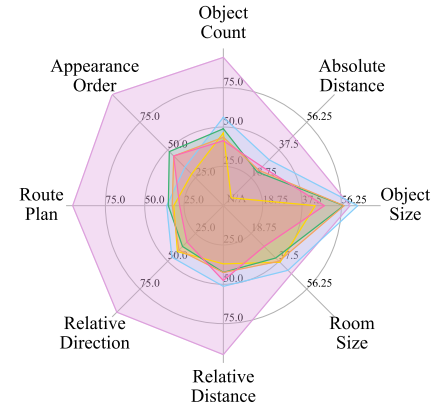


Table 1. **Evaluation on VSI-Bench. Left:** Dark gray indicates the best result among all models and light gray indicates the best result among open-source models. † indicates results on VSI-Bench (tiny) set. **Right:** Results including the top-3 open-source models.

- *Chance Level (Random)* is the random selection accuracy for MCA tasks (and is inapplicable for NA tasks).
- *Chance Level (Frequency)* represents the highest performance MLLMs would achieve by always selecting the most frequent answer for each task. This identifies performance gains that may result from inherently long-tailed answers or imbalanced multiple-choice distributions.

Human Level Performance. We sample a subset of 400 questions (50 per task), which we will refer to as VSI-Bench (tiny). Human evaluators independently answer each question, and their performance is evaluated using the above-mentioned metrics. For comparison, we also report Gemini-1.5 Pro's performance on VSI-Bench (tiny). See Appendix C for details on evaluation setups.

4.2. Main Results

Tab. 1 shows overall model performance on VSI-Bench. Our key observations are as follows:

Human Level Performance. Not surprisingly, human evaluators achieve 79% average accuracy on our benchmark, outperforming the best model by 33%. Notably, human performance on configuration and spatiotemporal tasks is remarkably high, ranging from 94% to 100%, indicating human intuitiveness. In contrast, the performance gap between humans and the best MLLM is much narrower on the three measurement tasks that require precise estimation of absolute distance or size, suggesting that MLLMs may have a relative strength in tasks requiring quantitative estimation.

Proprietary MLLMs. Despite a significant performance gap with humans, the leading proprietary model, Gemini-1.5 Pro, delivers competitive results. It surpasses the chance level baselines by a substantial margin and manages to approach human level performance in tasks such as absolute distance and room size estimation. It's worth noting that while human evaluators have years of experience in understanding the physical world spatially, MLLMs are only trained on 2D digital data like internet videos.

Open-source MLLMs. Top-tier open-source models like LLaVA-Video-72B and LLaVA-OneVision-72B demonstrate highly competitive performance to closed-source models, trailing the leading Gemini-1.5 Pro by only 4% to 5%. However, the majority of open-source models (7/12) perform below the chance level baseline, indicating significant limitations in their visual-spatial intelligence.

5. How MLLMs *Think in Space* Linguistically

To better understand when and why models succeed or fail and to elucidate the facets of visual-spatial intelligence they possess, we examine how MLLMs *think in space* linguistically here and visually in Sec. 6. We begin by prompting the best-performing MLLM in VSI-Bench, Gemini-1.5 Pro [78], to articulate its internal reasoning in language.

5.1. Probing via Self-Explanations

Self-explanations are a prevailing approach on par with traditional model explanations like LIME saliency maps [71] for understanding LLM-generated responses [25, 32, 53] and are widely used in analyzing language model behavior [66, 99]. We randomly sample a subset of 163 incorrect

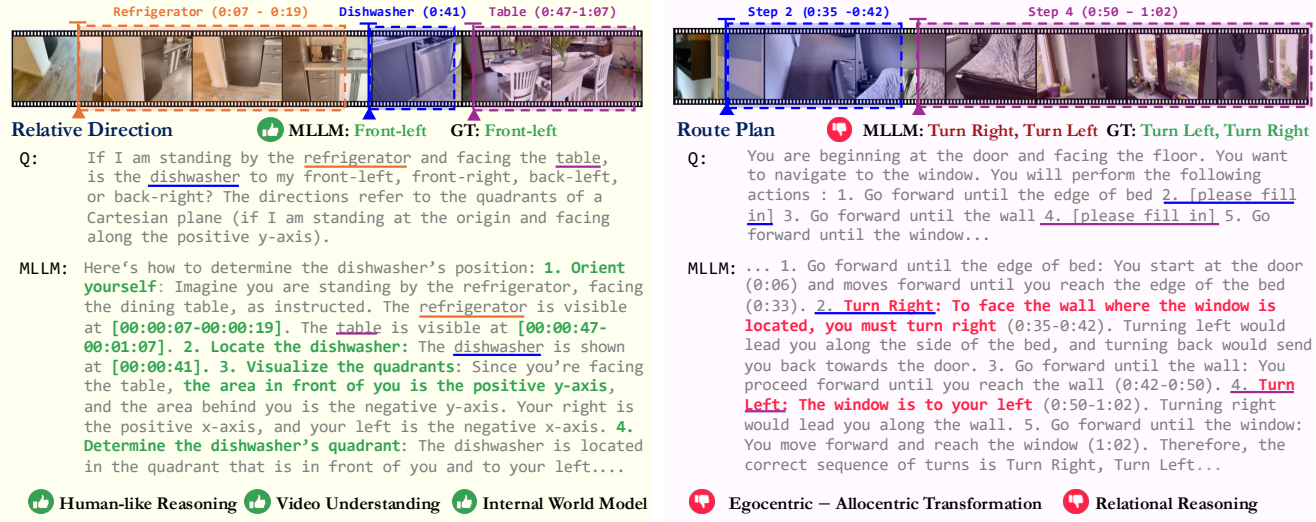


Figure 6. **Examples of how a MLLM thinks as seen in self-explanations**. While a MLLM exhibits strong video understanding and linguistic reasoning capabilities, its spatial reasoning capabilities are still developing.

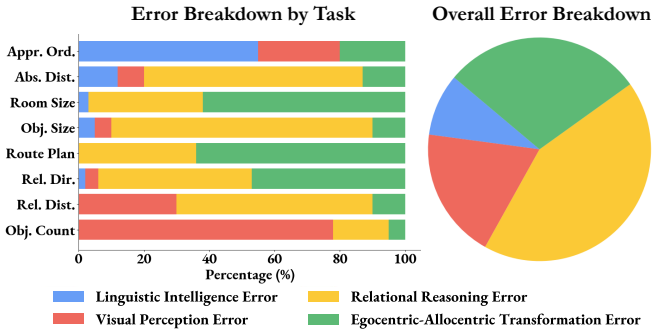


Figure 7. **Human-conducted analysis of errors by type.** Over 70% of errors stem from faulty spatial reasoning capabilities.

answers, prompt the MLLM to provide explanations for the predicted answers, and carefully review them by hand.

Case Studies. Fig. 6 presents self-explanations in both a success and an error case. In both examples, when thinking in space, the MLLM exhibits advanced video understanding, demonstrated by the impressive accuracy of its timestamped descriptions. The model also forms correct step-by-step reasoning processes, outlining steps such as "orient yourself", "locate the dishwasher" and "visualize the quadrants" for the relative direction task. Furthermore, the construction of a global coordinate system (Fig. 6, left) suggests that MLLMs may possess or build an implicit world model. Rather than using isolated frames, short clips, or random guesses, the MLLM used global spatial context and reasoning to infer correctly.

In the incorrect example (Fig. 6, right), we can identify faulty visual-spatial capabilities like egocentric-allocentric transformation and relational reasoning, as introduced in Fig. 2. In the video, the camera pans right to shift the view from the edge of the bed to the wall and window. The model obeys this egocentric view, responding that "to face the wall where the window is located, you must turn right" instead of creating an allocentric view reflecting the reality that the route from door to bed means turning left.

Error Analysis. To quantify and identify the main bottleneck for the best-performing MLLM on our benchmark, we analyze its errors on `VSI-Bench` (tiny), categorizing them into four distinct types which arose from both our outlined visual-spatial capabilities (Fig. 2) and a clear four-way categorization of errors upon examination:

1. **Visual perception error**, stemming from unrecognized objects or misclassified object categories;
2. **Linguistic intelligence error**, caused by logical, mathematical reasoning, or language understanding defects;
3. **Relational reasoning error** includes errors in spatial relationship reasoning, *i.e.*, distance, direction, and size;
4. **Egocentric-allocentric transformation error**, resulting from an incorrect allocentric spatial layout or improper perspective-taking.

As shown in Fig. 7, around 71% of errors are attributed to spatial reasoning (as ontologically conceived in Fig. 2), which suggests that:

> *Spatial reasoning is the primary bottleneck for MLLM performance on `VSI-Bench`.*

Further analysis and case studies are in Appendix E.2.

5.2. Limits of CoT Methods in Visuospatial Tasks

Prompting techniques improve the reasoning and problem-solving abilities for large models across diverse tasks [33, 35, 75, 84]. Their successes motivate us to investigate whether these linguistic prompting methods could also improve the visual-spatial capabilities of MLLMs in `VSI-Bench`. We investigate three prevailing prompting techniques (see Appendix B.3 for more details):

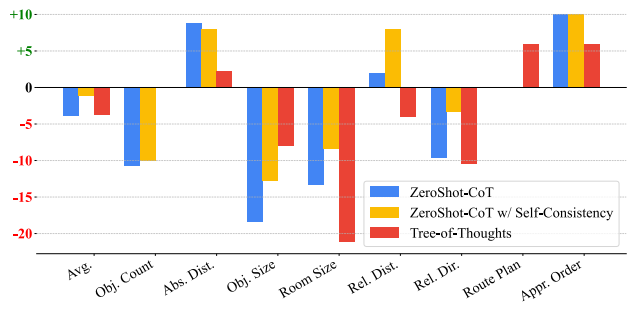


Figure 8. Relative improvements of *CoT*, *self-consistency* and *Tree-of-Thought* compared to the baseline. All three prevailing prompting techniques fail on average on our benchmark, and, in some cases, task performance becomes *much worse* after applying them. This implies that `VSI-Bench` cannot be solved by solely improving linguistic capabilities.

Case	Performance
Gemini-1.5 Pro (w/o CoT)	77.2
Gemini-1.5 Pro (w/ CoT)	79.8

Table 2. **Gemini-1.5 Pro CoT performance on a 500-questions subset in VideoMME.**

- *Zero-Shot Chain-of-Thought (CoT).* Following [38, 89], we add "Let's think step by step" to the prompts.
- *Self-Consistency w/ CoT.* We follow [87] and set the MLLM's temperature to 1.0 to encourage diverse reasoning and then take the majority consensus from five runs (employed w/ Zero-Shot CoT) as the final prediction.
- *Tree-of-Thoughts (ToT).* Following the "Creative Writing" practice in [95], we divide reasoning into plan generation and answer prediction. The MLLM first drafts and selects a plan, then generates three candidate answers and selects the most confident one as prediction.

As shown in Fig. 8, surprisingly, all three linguistic reasoning techniques lead to performance degradation on `VSI-Bench`. Zero-Shot CoT and ToT reduce average performance by about 4%, and self-consistency, though slightly better, still falls 1.1% below the no-prompting baseline. The unilateral improvement in the appearance order and absolute distance estimation tasks is easily explained by their significant percentage of linguistic intelligence errors (see Fig. 7). In contrast, the room size and object size tasks suffer a large 8% to 21% decrease, showing that encouraging a model to think more can be not just unreliable but downright harmful. Meanwhile, as shown in Tab. 2, ZeroShot CoT achieves a 1.6% improvement on the general video understanding benchmark VideoMME [24]. Therefore, our results suggest that:

Linguistic prompting techniques, although effective in language reasoning and general visual tasks, are harmful for spatial reasoning.

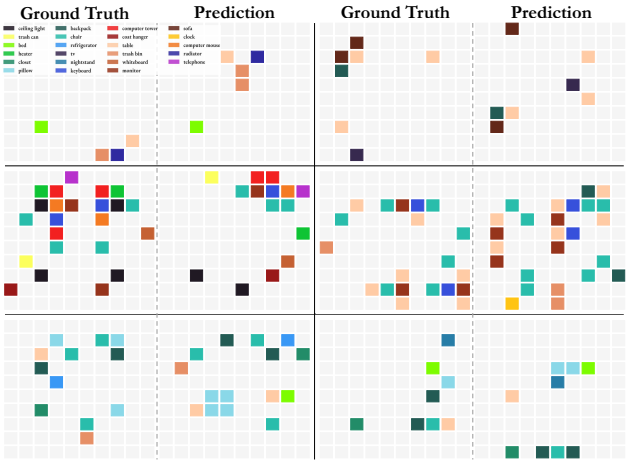


Figure 9. Visualization of cognitive maps from MLLM and GT.

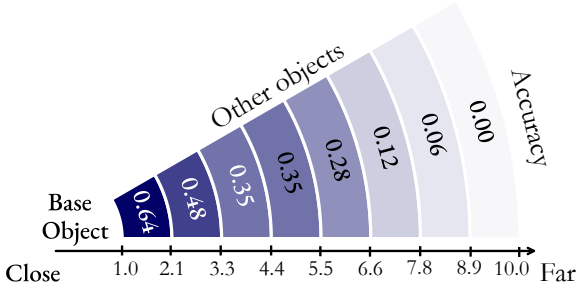


Figure 10. **Locality of the MLLM's predicted cognitive maps.** The MLLM's map-distance accuracy decreases dramatically with increasing object distance.

6. How MLLMs *Think in Space* Visually

Since humans subconsciously build mental representations [60, 80] of space when reasoning spatially, we explore how MLLMs remember spaces.

6.1. Probing via Cognitive Maps

We prompt MLLMs to express their internal representations of the spaces they see using cognitive maps, a well-established framework for remembering objects in a set environment [62, 80]. We prompt the best-performing MLLM, Gemini-1.5 Pro, to predict object center positions within a 10 × 10 grid based on video input (see Fig. 10b for grid size ablation and Appendix B.4 for prompt). We present examples of the resulting cognitive maps in Fig. 9.

To quantitatively assess these cognitive maps, we evaluate the Euclidean distance between all pairs of objects within each map. We consider the distance (on the grid) between two objects to be correct if it deviates by no more than one grid unit from the distance in the ground truth cognitive map. As shown in Fig. 10, we divide the map-distances into eight distinct bins for analysis. Interestingly, we find that the MLLM achieves a remarkable 64% accuracy in positioning adjacent objects within its cognitive map, indicating robust *local* spatial awareness. However, this accuracy significantly deteriorates as the distance between two objects increases, which suggests that:

When remembering spaces, a MLLM forms a series of local world models in its mind from a given video, rather than a unified global model.

This observation aligns with the challenge of forming a global space representation from discrete video frames, which is inherently difficult for MLLMs. While this task may not be trivial for humans either, it is likely that they can build such global space representations more accurately.

6.2. Better Distance Reasoning via Cognitive Maps

Given the local awareness of MLLMs in remembering spaces (see Fig. 9 and Fig. 10) and the importance of mental imagery to how humans think in space, we investigate whether generating and using cognitive maps can help MLLMs' spatial reasoning in terms of VSI-Bench's relative distance task. This tests if the local distance awareness emerged through cognitive maps transfers to improved distance recall and reasoning.

Case	Rel. Dist Acc.
w/o Cog. map	46.0
w/ Cog. map	56.0
w/ Cog. map (GT)	66.0

(a) Cognitive map prompting.

Cog. Map Src.	Size	Rel. Dist Acc.
MLLM	10×10	56.0
MLLM	20×20	54.0
GT	10×10	66.0
GT	20×20	78.0

(b) Cognitive map canvas size.

Table 3. **Relative distance task with cognitive map.**

We prompt Gemini-1.5 Pro to first generate a cognitive map based on the given video and question, and then to use the predicted map to answer the question. As shown in Tab. 3a, we find that using mental imagery improves a MLLM's relative distance accuracy by 10%. The 20% to 32% gain over baseline with the ground truth cognitive map underscores the importance of building accurate mental maps of a scene, which enforce a globally consistent topology, but indicates that such mental imagery is only one part of the puzzle, albeit a crucial one. These results point to building a mental spatial world model or cognitive map as a valuable pretext task or a promising solution for MLLMs to tackle visual-spatial reasoning.

7. Related Works

Apart from visual-spatial intelligence in Sec. 2, we further ground our work in the following two related areas:

MLLMs with Visual-Spatial Awareness. Building on the powerful language and reasoning abilities of LLMs [3, 9, 67, 68, 77, 81, 82] and the feature extraction abilities of modern vision encoders [30, 65, 69], MLLMs, especially visual MLLMs, exhibit unprecedented visual understanding capabilities [34, 40, 78, 86, 91, 102, 103], a promising direction toward developing world models [50] and embodied agents [17, 21, 37, 59]. However, grounding MLLMs in the real world presents significant challenges for models' visual-spatial intelligence, motivating recent efforts [10, 13, 16, 29, 41, 48, 94, 107]. Unlike prior works, which primarily focus on understanding spatial information through 2D images [70, 76, 93] or solely language [58, 72, 90, 90, 92], our work assesses models' visual spatial intelligence using real-world videos, which more closely mirrors human understanding of the world and application scenarios for embodied agents.

Benchmarking MLLMs on Video. With MLLMs displaying impressive performance on still-images across perception, reasoning, and multi-disciplinary tasks [39, 52, 98, 99], there is increasing interest in evaluating MLLMs' video understanding capabilities [23, 24, 43, 44, 47, 51, 54, 55, 63, 85, 96]. For example, Video-MME [24] comprehensively evaluates MLLMs across various video-related tasks, including recognition and perception. EgoSchema [55] and OpenEQA [64] evaluate MLLMs' understanding abilities using egocentric videos. Despite their significance, most prior works focus on content-level understanding [24, 43, 55, 63], which primarily serves as a temporal extension of 2D image understanding without 3D spatial consideration. Extending beyond prior benchmarks, our work focuses on spatial intelligence and requires core spatial capabilities like visual working memory and implicit scene reconstruction.

8. Discussion and Future Work

We study how models see, remember, and recall spaces by building VSI-Bench and investigating the performance and behavior of MLLMs on it. Our analysis of how MLLMs think in space linguistically and visually identifies existing strengths (*e.g.*, prominent perceptual, temporal, and linguistic abilities) and bottlenecks for visual-spatial intelligence (*e.g.*, egocentric-allocentric transformation and relational reasoning). While prevailing linguistic prompting methods fail to improve spatial reasoning, building explicit cognitive maps does enhance the spatial distance reasoning of MLLMs. Future avenues of improvement include task-specific fine-tuning, developing self-supervised learning objectives for spatial reasoning, or visuospatial-tailored prompting techniques for MLLMs.

Acknowledgments. We thank Ellis Brown, Ryan Inkook Chun, Youming Deng, Oscar Michel, Srivats Poddar, Xichen Pan, Austin Wang, Gavin Yang, and Boyang Zheng for their contributions as human annotators and evaluators. We also thank Fred Lu for proofreading our manuscript. We also thank Chen Feng, Richard Tucker, Noah Snavely, Leo Guibas, and Rob Fergus for their helpful discussions and feedback. This work was mainly supported by the Open Path AI Foundation, Google TPU Research Cloud (TRC) program, and the Google Cloud Research Credits program (GCP19980904). SX also acknowledges support from Intel AI SRS, IITP grant funded by the Korean Government (MSIT) (No. RS-2024-00457882, National AI Research Lab Project), Amazon Research Award, and NSF Award IIS-2443404.

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